

1. What are the key differences between OSPF and RIP? (3 marks)
2. Describe the OSPF Hello Protocol and its significance in neighbor discovery and network stability (4 marks)
3. Explain the role of BR and BDR in OSPF. (3 marks)
4. Explain the OSPF routing Algorithm. (10 marks)

Institute of Information Technology

Jahangirnagar University

3rd Year 1st Semester B.Sc (Hons.) in ICT (50 Batch)

Tutorial: 02

Course: ICT-3107 (Information & Data Security)

Time: 40 Minutes

Full Marks: 20

Answer any TWO of the following questions.

- 1a) Define cryptography. List some widely used cryptosystem. (5)
- b) State four security needs provided by cryptography. (5)
- 2a) Give a general idea behind symmetric-key cryptography. (5)
- b) How does asymmetric-key cryptography provide the message confidentiality (5)
- 3a) What are the merits and demerits of asymmetric-key cryptography over symmetric-key cryptography? (5)
- b) Three pass protocol can be used to send sensitive information across an insecure network. Give a postal analogy. (5)

Institute of Information technology
Jahangirnagar University
ICT 3101 – Operating System
Class Test :01

Marks: 10

Time: 30 minutes

1. What is the purpose of interrupts? How does an interrupt differ from a trap? Discuss the interrupt-driven I/O cycle with figure. 03
2. Identify several advantages and several disadvantages of open-source operating systems. Identify the types of people who would find each aspect to be an advantage or a disadvantage. 02
3. What is Direct Memory Access (DMA), and how does it improve system performance by reducing CPU intervention during data transfers between high-speed I/O devices and main memory? 02
4. What is the main advantage of the microkernel approach to system design? How do user programs and system services interact in a microkernel architecture? What are the disadvantages of using the microkernel approach? 03

Marks: 10

Time : 35 minutes

1. Describe different pieces of information associated with the PCB. Draw the figure how cpu switches from process to process. Illustrate two models of IPC. 04
2. What are two differences between user-level threads and kernel-level threads? Under what circumstances is one type better than the other? 02
3. Consider the following set of processes, with the length of the CPU burst time given in milliseconds. 04

Process	Burst Time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.

- a. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).
- b. What is the turnaround time of each process for each of the scheduling algorithms in part a?
- c. What is the waiting time of each process for each of these scheduling algorithms?
- d. Which of the algorithms results in the minimum average waiting time (over all processes)?

Marks: 10

3. A small computer lab has 4 computers and 3 types of resources (RAM, CPU cores, Storage). Time : 30 minutes

05

	Allocation Matrix			Max Matrix			Available Matrix		
	RAM(GB)	CPU Cores	Storage(GB)	Max RAM	Max CPU	Max Storage(GB)	RAM(GB)	CPU Cores	Storage(GB)
P1	3	2	50	6	4	100	8	6	150
P2	2	1	30	4	2	60			
P3	1	3	20	3	5	40			
P4	4	2	40	5	3	80			

- d. Determine if the system is in a safe state
- e. Create a safe sequence of resource allocation
- f. Verify if all processes can be completed without deadlock.

4. Explain Lock variable mechanism . Does Lock variable met the four conditions of Synchronization mechanism ? Explain, how this problem can be solved ?

05

ICT 3101 – Operating System
Class Test :04

SET -B

Marks: 10

Time: 25 minutes

1. What is page fault? Discuss the steps in handling a page fault? What Happens if there is no Free Frame? 04
2. How many page faults occur for First In First Out (FIFO) and **Least Recently Used (LRU) algorithm** for the following reference string with four-page frames? What will be the hit ratio?
1 2 3 4 5 3 4 1 6 7 8 7 8 9 7 8 9 5 4 5 4 2 04
3. Explain SSTF with its advantages and disadvantages. 02

Jahangirnagar University
Department of IIT, Course # IIT 3109(Basic-Economics)
2nd Tutorial/Midterm Examination-2024, 50 batch
Marks: 25, Time: 1.20 hours

Answer any one(01) of the following:

1.a) Given a demand equation is $Q_d = 40 - 4P$; where Q_d is the quantity demand and P is the price and a supply equation is $Q_s = 20 + 2P$; where Q_s is the quantity supplied and P is the price. Draw the demand and supply curve correctly and show the market equilibrium point and calculate the equilibrium price and quantity. 8

b) What will happen on the demand curve if (i) Consumer's income will increase ,(ii) Number of family members will decrease . Explain with graphs. 6

c) What will happen on the supply curve if (i) The price of raw materials will increase, (ii) The technological innovation will occur in the economy. Explain with graphs. 6

d) Define the price elasticity of software demand with formula. Draw the curves for perfectly elastic, perfectly inelastic, unitary elastic, relatively elastic and relatively inelastic demand. 5

2.a) Define market equilibrium. Suppose, A hypothetical market is in equilibrium position- If the market price will increase and will decrease from the equilibrium price, what will happen on the economy-explain (shortage and surplus) with figure. 10

b) Use the demand and the supply diagram, what will happen in the economy? 8

i) If the consumer's income will decrease and raw material price will increase.

ii) If the consumer's preference will increase and the no. of sellers will increase in the economy.

c) If the computer related commodity price is 20 units and quantity is 50 units. When price will 10 unit and quantity will 65 unit. Draw the demand curve and calculate price elasticity of demand. 7

Jahangirnagar University
Department of IIT, Course # IIT 3109(Basic-Economics)
1st Tutorial Examination-2024, 50 batch
Marks: 25, Time: 50 minutes

2 X 9 = 18

Answer of the followings:

1. What is PPF? Draw a figure of PPF. Show the efficient, inefficient and unattainable point on PPF and why?
2. What is the meaning of demand in economics? Describe the law of demand and prove the demand law with figure. Difference between change in demand and change in quantity demanded with figure.

0.5 X 14 = 7

Answer the following MCQs:

1. Society has limited resources and therefore cannot produce all the goods and services people wish to have, which means- A. Scarcity B. Resources C. Tradeoff D. Price
2. Benefits of using resources are distributed fairly among society's members is called- A. Efficiency B. Equity C. Scarcity D. Resources
3. Whatever must be given up to obtain some item is called- A. Fixed cost B. Sunk cost C. Opportunity cost D. Variable cost
4. Which factor influences (shifts) the demand curve.? A. Income B. Resources C. Tradeoff D. Input prices
5. Which factor influences (shifts) the supply curve? A. Income B. Equity C. Tastes D. Technology
6. If income increases, then the consumption of a good also increases we call - A. Shift in the demand B. Changes in the quantity demanded C. Inferior good D. Changes in supply
7. If the raw material price decreases, what happen on the economy? A. The demand will increase B. The supply decrease C. The supply will increase D. No change

8. If the price of a good is fixed but the taste of a good increase. What will happen? A. Price increased and supply increased also B. Price decreased and demand increased also C. Price fixed and demand fixed D. Price fixed and demand increased

9. The determinant of demand is- A. Weather condition B. Supply schedule C. Future expected prices D. Resource price

10. Which one is expressed the income in economics -

A. C B. I C. Y D. S

11. If the price of a commodity increases, and the demand of other good also increases, we call it-

A. Substitute good B. Complementary good C. Cross good D. Luxuries good

12. If the price of a commodity changes, and the demand of same good also increase, we call it-

A. Substitute good B. Complementary good C. Cross good D. Luxuries good

13. The production will be efficient when the production point is

A. On the PPF B. Under the PPF C. Above the PPF D. Above all

14. The basic three questions of micro economics are -

i. What to produce ?

ii. How to produce ?

iii. -----.

Jalangi Nagar University
Department of IIT, Course # IIT 3109(Basic-Economics)

3rd Tutorial Examination-2024, 50 batch

Marks: 25, Time: 55 minutes

Answer any one of the followings:

1. a) Draw and briefly explain the circular flow diagram. Identify the parts of the model that correspond to the flow of goods and services and the flow of taka for each of the following activities:

$12+5+8=25$

- i) Rahi pays a store keeper tk.110 for a quart of milk.
- ii) Bintu earns tk.100 per hour working at a first food restaurant.
- iii) Rimu spends tk.500 to see a drama at the theatre house in Green Road.
- iv. Arif earns tk.10,000 from his 10 percent ownership of Acme Industrial.

b) Why does the aggregate demand (AD) curve slopes downward? Explain any one reason.

c) Explain with appropriate figures what happen of price level and output level through aggregate demand (AD) curve :

i. If investment expenditures decrease

ii. If Govt. expenditures increase

2.a) What are the differences between GDP deflator and ^{GNP} consumer price index (CPI)?

$6+12+7=25$

b) Below are some data on three goods produced in the economy.

Year	Price of rice(/kg) in Tk.	Quantity of rice (kg)	Price of cloth (/yard) in Tk.	Quantity of cloth (/yard)	Price of flour(/kg) in Tk.	Quantity of flour (kg)
2015	5	100	10	250	12	150
2016	10	200	20	300	15	180
2017	20	300	25	350	22	250

- i) Compute nominal GDP, real GDP, GDP deflator and economic growth rate for each year, using 2015 as the base year.
- ii) Explain the economic meaning of above results.
- c) Prove that GDP's income is equal to GDP's expenditure with graph.